

SYNOPSIS



(All times are UTC)

At about 0408 on 23 January 2005, just south of Grimsby Middle in the River Humber, the UK registered 1696gt tanker *Amenity* collided with the Norwegian registered ro-ro cargo vessel *Tor Dania*. Both vessels suffered significant damage but there were no injuries or pollution and both vessels were able to continue to berth un-aided before being withdrawn from service for repairs. There was a north-westerly wind blowing force 4 to 5, visibility was good and there were moderate seas.

Both vessels were being piloted by their respective masters, who held Pilotage Exemption Certificates (PECs) for the Humber. *Amenity* was outbound from Immingham Oil Terminal (IOT) with a cargo of 815t petrol and 1435t diesel. *Tor Dania* was inbound from Cuxhaven with four passengers and a cargo of new cars.

As *Tor Dania* made her turn for the South Shoal buoy, south of Grimsby Middle, the master of *Amenity* incorrectly concluded that *Tor Dania* had turned onto a collision course. He decided that his only option was to put the engine of his vessel full astern and, in an attempt to counter the port swing that this would induce, he put the steering hard to starboard. However, *Amenity* turned to port and hit *Tor Dania* close to midships on the port side at a speed of about 7 knots.

There was no ship's machinery failure or influence from the prevailing conditions; the collision occurred as a result of the actions taken by *Amenity*'s master. He expected *Tor Dania* to display a green sidelight briefly, as she turned for the South Shoal buoy at Clee Ness. However, it is likely that he was either distracted or mentally overloaded, perhaps by completing the 0400 logbook entries, and when he looked back up he was not presented with the sight he was expecting. This might have led to him making a quick decision based on a perceived emergency situation. However, *Amenity* had sufficient depth of water to navigate safely outside the main channel, and ample space to manoeuvre out of the way without reducing speed.

ABP is restricted in the training and examination requirements it can impose on applicants for PECs; they may not be more onerous than those applied to pilots. The training the master of *Amenity* received to qualify for his PEC consisted of many voyages inbound under the supervision of both another PEC holder and authorised pilots, in addition to the study required for the examination. He had not completed any simulator training.

Under the Port Marine Safety Code (PMSC), each Competent Harbour Authority (CHA) is allowed to prescribe its own requirements for the authorisation of pilots and PEC holders. This leads to some CHAs requiring a practical stage to the PEC examination on board the vessel and/or in a vessel simulator, while others require only an examination of the theory, using table-top exercises.

The MAIB investigation identified the following safety issues:

- According to the local bye-laws, *Amenity* was the give way vessel as she was sailing against the tide.
- The master of *Amenity*:
 - Might have become overloaded in piloting the vessel.
 - Made a quick decision based on scanty information, and did not re-evaluate this decision as more information became apparent.
 - Took action that was contrary to rule 7c of the collision regulations.
 - Had no other officer present on the bridge to assist him and monitor the pilotage.
 - Did not use his lookout to best effect.
 - Did not have a formally assessed act of pilotage as part of his PEC training or examination.
 - Displayed poor practical pilotage abilities in an emergency situation that were not recognised during the training and examination for his PEC.
- There are no national standards for the examination and issuing of PECs to applicants. The PMSC requires CHAs to ensure best practice in addressing the risks identified in their waters.

Following this accident:

- Associated British Ports Humber Estuary Services (ABP HES) took a number of actions. These included altering their procedures so that each PEC candidate is either formally assessed by a senior pilot on his final qualifying trip, or the examination is conducted on passage.
- F.T. Everard has also taken a number of actions, including implementing a new bridge procedure throughout the fleet. The new procedure requires two qualified navigating officers to be present on the bridge at all times when in pilotage waters, except in the area immediately off the berth.

The MAIB has recommended that the Port Marine Safety Code Steering Group evaluate, then promulgate current industry, best practice to port operators on the issuance of PECs, in the form of guidelines. Such guidance should recognise the need for:

- Practical evaluation of the PEC candidate's local knowledge and ship-handling ability.
- Assessment of the candidate's ability to cope with foreseeable emergency and/or high density traffic scenarios.
- Verification of the relevant bridge team manning arrangement, to ensure appropriate levels of support for the PEC holder during port movements.

SECTION 1 - FACTUAL INFORMATION

1.1 PARTICULARS OF VESSELS AND ACCIDENT

Vessel details

Amenity (Figure 1)

Registered owner	:	Everard Shipping
Manager(s)	:	F.T.Everard & Sons Ltd.
Port of registry	:	London
Flag	:	UK
Type	:	Tanker
Built	:	10/1980
Classification society	:	Bureau Veritas
Length overall	:	79.23 m
Gross tonnage	:	1696gt
Engine power and/or type	:	Ruston 12 Cyl. 2000 HP reversing gearbox
Propulsion	:	Single fixed pitch propeller
Service speed	:	13.4 knots
Other relevant info	:	Bow thrusters

Vessel details

Tor Dania (Figure 2)

Registered owner	:	Seaheron A/S
Manager(s)	:	Goliath Shipping A/S
Port of registry	:	Oslo
Flag	:	Norway
Type	:	Ro-Ro container
Built	:	1978
Classification society	:	Bureau Veritas
Length overall	:	193.25 m
Gross tonnage	:	21850
Engine power and/or type	:	Sulzer 12cyl 17400 HP
Propulsion	:	Single controllable pitch propeller
Service speed	:	19 knots
Other relevant info	:	2 bow and 1 stern thruster, Becker rudder.

1.2 BACKGROUND

1.2.1 *Amenity*

Amenity (**Figure 1**), was a 1696grt oil products tanker. Her home berth was Immingham Oil Terminal (IOT) in the Humber, and from there she sailed to UK ports.

Passages were between 12 and 30 hours, depending on the destination, and the vessel was normally alongside IOT for about 12 hours. She carried three bridge watchkeeping officers, including the master. The master would take the vessel in and out of port, while the chief officer was responsible for the discharge of cargoes and the second officer for loading cargoes. At sea, the watches were arranged to ensure that the officer required for cargo duties on arrival at the next port, was fully rested.

The vessel also carried three ABs, two remained on 6 on/6 off watches and were used on the bridge as lookouts during the hours of darkness.

Under normal conditions, *Amenity* had sufficient depth of water to enter IOT from 1 hour after low water.

The vessel could maintain steerage down to a speed of 2-3 knots. She had a left-handed fixed pitch propeller, meaning that the bow would swing to port when the propeller was going astern (**Figure 3**).

The vessel's draught at the time of the accident was 5.6m. It was the master's practice to leave a minimum of 0.8m under keel clearance.

1.2.2 *Tor Dania*

Tor Dania (**Figure 2**) was a 21491grt ro-ro cargo vessel operating between Cuxhaven, Germany and Immingham Dock. Weather permitting, the vessel would complete three round trips every week.

Passages were normally about 21 hours at 17 knots, and the vessel was usually alongside for 6 to 7 hours at both ports. *Tor Dania* carried a master, chief officer and two second officers who worked 6 on/6 off watches at sea, with lookouts posted at all times.

At the time of the accident, the bulk of the cargo being carried was new cars being exported to the UK. These were loaded and discharged by the car's manufacturer.

Tor Dania was not tidally constrained for Immingham Dock.

The vessel's draught at the time of the accident was 7m. It was the master's practice to maintain at least 3m under the keel when *Tor Dania* was proceeding at speed greater than 12 knots.

1.2.3 Associated British Ports Humber Estuary Services (ABP HES)

The Pilotage Act 1987 made Associated British Ports (ABP) the Competent Harbour Authority for the Humber at the time of the accident. However, many of the duties and responsibilities of ABP were delegated to Associated British Ports Humber Estuary Services (ABP HES), whose remit was to provide an efficient pilotage service.

This was achieved through the issue, last revised in July 2004, of pilotage directions for ships to be navigated within the Humber pilotage area (available at <http://www.humber.com/statutory/index.asp>). One of the requirements of these directions was that all vessels of 60m and above, and all vessels carrying oil or hazardous cargoes in bulk, were to be in the charge of a pilot or a Pilotage Exemption Certificate (PEC) holder when in the pilotage area.

Where the PEC was used instead of taking a pilot, the dues paid to ABP were reduced to one quarter of those applicable if a pilot had been carried. Additionally, these fees were only due for the first 80 acts of pilotage per year, and thereafter the PEC holder would not be required to pay any pilotage dues.

Three classes of PEC were available, each with their own assessment requirements. Class A covered vessels over 100m, Class B vessels under 100m and Class C for river craft greater than 20m in length not passing to or from sea but carrying dangerous goods in bulk.

The pilotage directions also specified the experience and knowledge that was expected of applicants for PECs.

The practical experience required of masters or chief officers applying for a class A or B PEC, consisted of evidence of at least nine voyages in and out of the port, during the preceding 18 months, under the supervision of a PEC holder or a pilot. The qualifying passages had to be undertaken in the part of the river for which the PEC related on board a vessel of a substantially similar class.

The general and local knowledge required for each class of PEC was also specified in the pilotage directions. All applicants were assessed, interviewed and examined by ABP HES, and every PEC was reviewed annually. Additionally, any applicant for a PEC could be required to have their practical competencies examined on board their vessel during a river transit. However, the PEC examination was not permitted to be more onerous than the Pilotage examination.

1.3 ENVIRONMENTAL CONDITIONS

At the time and position of the accident, there was a north-westerly wind blowing force 4 to 5, visibility was good and there were moderate seas.

23 January was three days after neap tides. High water was 6.3m above chart datum at 0453 in Hull, 0440 in Immingham and 0431 in Grimsby. In the area of Grimsby Middle, the tide was setting to the north-west by $\frac{1}{2}$ to $\frac{3}{4}$ knots.

1.4 NARRATIVE

All times are UTC, all courses true and all speeds are speed over the ground (SOG).

At 0301 on the morning of 23 January 2005, *Amenity* sailed from the Immingham Oil Terminal (IOT) for Portland, Dorset with the master and one AB lookout on the bridge. She had a draught of 5.6m and was loaded with unleaded petrol and diesel.

The master engaged the autopilot once clear of the berth. He was navigating visually, and observing the Electronic Chart System (ECS) and the radar, which was set to 3 miles range. He was following the previously drawn passage plan on the paper chart, marking the position as the vessel passed the buoys (**see Figure 4 - the chart in use on *Amenity***).

An AB was on the bridge as a lookout in accordance with company instructions. However he was not involved by the master in monitoring progress of *Amenity* or other vessels in the vicinity.

The master was acquiring radar targets ahead to monitor their speed and heading and the risk of collision. However, he was not overlaying radar targets on the ECS or making use of the available Automatic Identification System (AIS) information. The ECS displayed the planned track and waypoints as shown in **Figure 5**, a screenshot taken from *Amenity*'s ECS taken after the accident.

At 0329, *Amenity*'s master reported to Vessel Traffic Services (VTS) Humber as he passed No.9A buoy.

At 0336, *Tor Dania*, inbound from Cuxhaven with a draught of 7.0m, passed Alpha Buoy. The master reported in to VTS, as required, and VTS reported little traffic, enabling *Tor Dania* to proceed directly to Immingham lock.

The officer of the watch (OOW), the master and an AB helmsman were on the bridge of *Tor Dania*. They were using two radars, one set to the 1.5 mile range and the other being switched between the 3 and 6 mile ranges by the OOW, to check the traffic ahead. The master was navigating visually, observing the radar set to the 1.5 mile range. He used its ARPA function to monitor the speed and courses of other vessels, and hence assess any risk of collision. The master was also referring to the ECS, but was judging *Tor Dania*'s position from visual references. The master did not make use of the ECS radar overlay function and he did not refer to the available AIS data.

At about 0355, *Tor Dania* was heading north-west in the Bull Channel when the master saw an outbound vessel at about 6 miles and acquired her with the ARPA. The outbound vessel was *Amenity*, 10° on his port bow and displaying a red navigational sidelight.

Figures 6 to 10 show representations of the position and aspect of both vessels on a chartlet of the area. *Amenity* is shown in red and *Tor Dania* in blue.

At 0400 (**Figure 6**), *Amenity*'s master saw an inbound vessel, displaying a red sidelight fine on the port bow at 3 miles. This was *Tor Dania*. At this time both vessels were navigating in approximately the centre of the buoyed channel.

Tor Dania's master reported to VTS when the vessel was passing abeam of Clee Ness No.4A light float at 0403 (**Figure 7**). At this time, he advised VTS that he was reducing speed to ensure his arrival at Immingham coincided with a break in the traffic there. He then reduced speed from 17 to 14 knots. *Amenity*'s master heard this and was, therefore, aware of the name of the inbound vessel.

At 0406 (**Figure 8**), *Tor Dania*'s master altered course to port to clear South Shoal buoy. *Tor Dania* overshot the turn fractionally, coming round to about 264° before adjusting course a little to starboard to pass close to the south of the South Shoal buoy. The vessel was to the north side of the channel, making good 14 knots. A snapshot taken from the vessel's ECS display is shown in **Figure 11a**.

Amenity's master expected to see *Tor Dania*'s green sidelight as she made her turn for South Shoal buoy. However, when he saw *Tor Dania* make her turn at about 0406, he interpreted what he saw as the vessel altering course to between 220° and 240°. As a laden tanker, he considered *Amenity* to be restricted to navigating within the buoyed channel. Therefore, faced with what he perceived as a vessel coming round across his bow with a closing speed of about 25 knots, he put his propulsion full astern. Aware that *Amenity* had a left-hand propeller, and that this manoeuvre would push the bow to port, he steered hard to starboard to counter the effect. However, as soon as *Amenity*'s propeller started to turn astern, the bow began to swing to port.

At 0407 (**Figure 9**), the master of *Tor Dania* noticed *Amenity* starting to alter course to port, and initially he thought that she was making her turn for the No.4B buoy. However, when the vessel continued to alter course, he realised that a collision was inevitable, and ordered the helm hard to starboard to reduce the angle of impact. The collision occurred about 30 seconds later (**Figure 10**), and at 0407 50 *Tor Dania*'s master reported the collision to VTS.

Amenity's master also reported the collision, some seconds later, initially referring to the other vessel as *Tor Hollandia*, before being corrected by *Tor Dania*'s master.

Despite a heavy impact, with *Amenity*'s bow making contact with the port side of *Tor Dania*, aft of midships, the vessels were able to continue, unaided, after having assessed the damage to both vessels.

SECTION 2 - ANALYSIS

2.1 AIM

The purpose of the analysis is to determine the contributory causes and circumstances of the accident as a basis for making recommendations to prevent similar accidents occurring in the future.

2.2 FATIGUE

Fatigue is not considered a contributory factor in this accident. Both masters had had ample opportunity to rest in the hours and days preceding the accident, and both claimed to have been fully rested.

2.3 THE COLLISION

There was no ship's machinery failure, or influence from the prevailing conditions. The collision occurred as a result of the actions taken by the master of *Amenity* when he incorrectly concluded that *Tor Dania* had turned onto a collision course. The possible reasons for this incorrect interpretation of *Tor Dania's* aspect, and the actions taken as a result, are discussed below.

2.3.1 *Tor Dania*

Tor Dania was generally being piloted near the centre of the channel, however, as the vessel approached the South Shoal buoy, she was positioned to the starboard side of the channel. There was therefore sufficient room available in the buoyed channel for *Amenity* to pass *Tor Dania* port to port.

Once the master saw that a collision was inevitable, he took appropriate evasive action. This action might have been taken earlier when he saw *Amenity* alter course to port and towards his own vessel. However, the alteration coincided with *Amenity's* expected navigational alteration to port around the South Shoal buoy, and the master had no reason to suspect that *Amenity* would not stop her swing to port and continue along the channel. It was only in the few seconds before impact, that it became apparent that *Amenity* was not going to stop the swing, and that impact was inevitable. Before taking evasive action, the master was aware that he had sufficient height of tide to clear Grimsby Middle bank.

In accordance with the recommendations contained in ABP Humber's General Notice to Pilots No 16/2004, the master of *Tor Dania* could have contacted the master of *Amenity* by VHF in order to confirm the other vessel's intentions. However, *Amenity* was not a large vessel, and *Tor Dania's* master saw no reason to be concerned at the prospect of passing a small vessel in the vicinity of the South Shoal buoy, as this was a situation he had encountered many times before.

2.3.2 *Amenity*

Leading up to the collision, *Amenity* was positioned towards the wrong side of the channel. The vessel had sailed up the channel on the starboard side, when inbound the previous day. However, outbound, the master kept the vessel to port of mid-channel. Pilots and PEC holders are instructed to keep to the starboard side of the channel unless agreement has been made to the contrary, although it is not uncommon for vessels to navigate towards mid-channel when clear of other traffic and to move over to the starboard side as required to leave sufficient space for other vessels to pass.

The relevant local bye-law's paragraph 15 (1) states:-

Subject to the provisions of paragraph (b) of this Byelaw, where a power-driven vessel is navigating against the tidal stream the master of the vessel shall, on approaching bends in the Humber or fairways or bridges, reduce speed or stop the vessel as necessary so as to allow any other vessel navigating with the tidal stream to pass clear of the vessel.

Therefore, *Amenity* was the give way vessel as she was sailing against the tide.

Amenity's master could give no explanation for his misinterpretation of the aspect of *Tor Dania*. He expected *Tor Dania* to show him a green sidelight briefly, as she turned for the South Shoal buoy at Clee Ness. However, he might have been momentarily distracted, perhaps by completing the 0400 logbook entries, and when he looked back up he was not presented with the aspect of the other vessel he was expecting. This might have led to him making a quick decision based on what he thought he saw.

Because he was the only officer on the bridge, *Amenity's* master was unable to verify his perception of the other vessel's intentions with another competent navigating officer. The presence of another officer might also have reduced the load placed on the master at this time, by removing the need for him to complete more routine tasks such as the completion of the log or making reports to VTS. It might also have allowed the master more time to maintain visual watch on the approaching vessel and to consider whether to contact *Tor Dania* in compliance with ABP Humber's General Notice to Pilots No 16.2004. *Amenity's* master claims to have checked the radar, and that this also indicated that *Tor Dania* was on a course of 220°-230°. Since *Tor Dania* had recently altered course and was now quite close, the ARPA vector on the radar would not have settled down to give reliable information, and could well have supported his mistaken visual impression of the vessel's aspect. Had the master referred to the available AIS information, he would have known *Tor Dania's* heading, as well as the direction and rate of turn. This information would have been updated at 2 second intervals, and could have corrected the mistaken visual impression.

In this perceived emergency situation, *Amenity's* master's decision to go astern is questionable. The decision was based on the fact that he thought that *Tor Dania* was now on a collision course and crossing his bows from port to starboard. His chart (**see Figure 4**) shows that he had hatched off all charted depths less than 5m, from the contour lines on both sides of the channel.

At the time of the accident, *Amenity's* draught was 5.6m aft. Company instructions required a minimum 10% underkeel clearance (ukc). In practice, the master stated that he would have maintained a ukc of 0.8m. However, he considered his vessel to be a fully laden tanker and, in his opinion, this meant that he had no option other than to stay within the buoyed channel as marked on his chart (**see Figure 4**). In reality, he had sufficient water at the time to safely navigate in charted depths close to datum, since the height of tide was estimated to be about 6m, and his draught was 5.6m. So if he was in a position where the charted depth was chart datum, 0m, he would have an under keel clearance of 0.4m. The charted depth at the point of the collision was about 8m, so the actual depth of the water at the point of the collision was about 14m. Just either side of the channel in that area, the charted depth was 7m. Therefore he had ample water to leave the buoyed channel.

The master was aware that, by putting *Amenity's* engine astern, her bow would swing to port (**see Figure 3**), and he attempted to counter this by applying full starboard helm. Having decided on this course of action, he did not feel that he could usefully do anything else, even when it became clear that *Tor Dania* was not on the course he had thought. However, it is clear from the electronic evidence, that he could only have seen a green sidelight on *Tor Dania* briefly, and would have seen the red sidelight again, at least 2 minutes before the collision. This should have given him sufficient time to put the engine ahead, and allow the vessel to turn to starboard, probably avoiding *Tor Dania* entirely.

The action taken by *Amenity's* master was based on a glance at the visual aspect of *Tor Dania*, possibly combined with the instantaneous information shown on the ARPA. The momentary visual aspect might have been observed as *Tor Dania* steadied on her new course after having slightly overshoot the turn to clear South Shoal buoy. *The International Regulations for Preventing Collision at Sea*, rule 7c, state that *assumptions shall not be made on the basis of scanty information, especially scanty radar information*. This rule was not complied with since his actions were based on scanty information.

2.4 AMENITY'S MANNING

The MAIB considered cultural issues with regard to the interaction between the master and the crew. However, there was no evidence of any issues in this regard.

ABs were always used as lookouts during the hours of darkness, and this was the case at the time of the accident. However, the AB on watch at the time of the accident was of limited value to the master because he was unaware of the vessel's position, so could not be expected to monitor or comment on the master's actions. In the absence of a second deck officer, the AB could have been better utilised to assist the master.

Amenity had sufficient complement to allow two deck officers to be on the bridge in pilotage waters. This would be dependent on compliance with the hours of work regulations, but based on *Amenity*'s movements over the days preceding the accident, this could have been arranged. In fact, the master had previously used another deck officer in pilotage waters, in conditions of heavy traffic or reduced visibility. However, this was neither the company's nor the master's standard practice, and the provision of an additional deck officer on the bridge had only been utilised on an ad-hoc basis.

In contrast to this, when a pilot is engaged on a vessel, he is in addition to the vessel's normal complement, and increases the bridge team by one. The other members of the bridge team are available to monitor the actions of the pilot and assist him as necessary.

Had there been another officer on *Amenity*'s bridge, monitoring the pilotage, it is possible that he would have been able to provide the master with a more considered assessment on the track of *Tor Dania*.

2.5 THE PILOTAGE EXEMPTION CERTIFICATE

In the PEC examination of *Amenity*'s master, carried out by ABP Humber, no onboard test or simulation training was required or undertaken. The examination was carried out by ABP Humber's Pilotage Operations Manager in October 2004. The Pilotage Operations manager was an experienced pilot and examiner, and his comments from this examination read as follows:

Captain [deleted] had a satisfactory knowledge of the river and its operation. He had obviously spent a good deal of time learning the buoys, currents and depths. Had a good explanation for all scenarios put to him.

However, following the accident, the Pilotage Operations manager could not explain or understand the master of *Amenity*'s actions as he approached *Tor Dania*.

Before the accident on 23 January 2005, *Amenity*'s master had made five previous departures in the dark as a PEC holder. Prior to examination, he had completed 38 qualifying trips with authorised pilots, none of whom had reported adversely on his competence, nor had they formally assessed him. The master had not completed any simulator training, although this was available through his shipping company.

ABP is restricted in the training and examination requirements it can impose on applicants for PECs; they may not be more onerous than those applied to pilots. Since all pilots were not required to undergo simulator training, ABP could not insist that this be done by PEC applicants. However, ABP could require PEC applicants to have their practical competencies examined on board their vessel during a river transit, in addition to the existing requirement to attend the examination centre.

Although pilotage examination includes discussion of emergencies, had the master's PEC assessment included an assessed act of pilotage, any weakness he might have had when dealing with emergency scenarios might well have been recognised. Additionally, it is likely that the way the master prepared his passage plan could have alerted the assessor to the fact that he was not making suitable use of the information available to him.

2.6 PORT MARINE SAFETY CODE

It is a requirement under the PMSC that the CHA keeps the need for pilotage services under review. This consideration should be part of the authority's hazard and formal risk assessment, and this should be formally re-assessed at no fewer than 3 yearly intervals.

The provision of a pilotage service is not discharged simply by authorising one or more pilots. It includes the management of the service to allow controls to ensure that the pilot assigned to a vessel is fit or appropriately qualified. It is for the CHA to determine the qualifications required of a pilot to become authorised for certain ship types and parts of the harbour.

The authorisation of PEC holders is included in the requirement for risk assessment, since they will form part of the pilotage service. Controls put in place to reduce the risks associated with PEC holders, could include a requirement for supervised acts of pilotage, limitation of the number of different ships authorised and the testing of local knowledge at an examination ashore.

Each CHA is allowed to prescribe its own requirements for the authorisation of pilots and PEC holders. This leads to some CHAs requiring a practical examination of the PEC candidate's capabilities on board the vessel and/or in a vessel simulator, while others require only to probe the candidate's competence in theory, using table-top exercises.

The PMSC has no national standards for qualifying trips/training of PEC holders and relies on the CHA to ensure best practice in addressing the risks identified in their waters.

The MAIB has made a recommendation to the Port Marine Safety Code Steering Group to evaluate and promulgate to CHAs current industry best practice on the issuance of PECs in the form of guidelines. These guidelines should recognise the need for assessment of a candidate's ability to cope with foreseeable emergency and/or high density traffic scenarios.

SECTION 3 - CONCLUSIONS

3.1 SAFETY ISSUES

The following are the safety issues which have been identified as a result of the MAIB's investigation. They are not listed in order of priority, but in the order in which they appear in Section 2.

1. According to the local bye-laws, *Amenity* was the give way vessel as she was sailing against the tide. [2.3.2]
2. The master of *Amenity* might have become overloaded while piloting the vessel. [2.3.2]
3. *Amenity*'s master made a quick decision based on scanty information, and did not re-evaluate this decision as more information became apparent. [2.3.2]
4. The action taken by *Amenity*'s master was contrary to rule 7c of the collision regulations. [2.3.2]
5. The master of *Amenity* had no other officer present on the bridge to assist him and monitor the pilotage. [2.4]
6. The master of *Amenity* did not use his lookout to best effect.
7. *Amenity*'s master's PEC training and examination did not include a formally assessed act of pilotage. [2.5]
8. *Amenity*'s master's poor practical pilotage abilities, in an emergency situation, went unrecognised during the qualifying trips and examination for his PEC. [2.5]
9. The PMSC has no national standards for qualifying trips/training of PEC holders, and relies on the CHA to ensure best practice in addressing the risks identified in their waters. [2.6]